

B11 Algebraic modelling

Name: _____

Class: _____

Date: _____

Time: **200 minutes**

Marks: **166 marks**

Comments:

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Q1.

The daily world production of oil can be modelled using

$$V = 10 + 100\left(\frac{t}{30}\right)^3 - 50\left(\frac{t}{30}\right)^4$$

where V is volume of oil in millions of barrels, and t is time in years since 1 January 1980.

- (a) (i) The model is used to predict the time, T , when oil production will fall to zero.

Show that T satisfies the equation

$$T = \sqrt[3]{60T^2 + \frac{162\,000}{T}}$$

(3)

- (ii) Use the iterative formula $T_{n+1} = \sqrt[3]{60T_n^2 + \frac{162\,000}{T_n}}$, with $T_0 = 38$, to find the values of T_1 , T_2 , and T_3 , giving your answers to three decimal places.

(2)

- (iii) Explain the relevance of using $T_0 = 38$

(1)

- (b) From 1 January 1980 the daily use of oil by one technologically developing country can be modelled as

$$V = 4.5 \times 1.063^t$$

Use the models to show that the country's use of oil and the world production of oil will be equal during the year 2029.

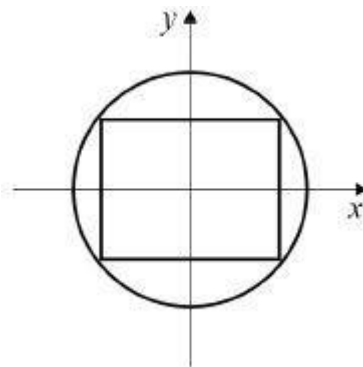
(4)

(Total 10 marks)

Q2.

A company is designing a logo. The logo is a circle of radius 4 inches with an inscribed rectangle. The rectangle must be as large as possible.

The company models the logo on an x - y plane as shown in the diagram.



Use calculus to find the maximum area of the rectangle.

Fully justify your answer.

(Total 10 marks)

Q3.

Sam goes on a diet. He assumes that his mass, m kg after t days, decreases at a rate that is inversely proportional to the cube root of his mass.

- (a) Construct a differential equation involving m , t and a positive constant k to model this situation.

(3)

- (b) Explain why Sam's assumption may not be appropriate.

(1)

(Total 4 marks)

Q4.

An open-topped fish tank is to be made for an aquarium. It will have a square horizontal base, rectangular vertical sides and a volume of 60 m^3 . The materials cost:

- £15 per m^2 for the base
- £8 per m^2 for the sides.

- (a) Modelling the sides and base of the fish tank as laminae, use calculus to find the height of the tank for which the overall cost of the materials has its minimum value.

Fully justify your answer.

(8)

- (b) (i) **In reality, the thickness of the base and sides of the tank is 2.5 cm**

Briefly explain how you would refine your modelling to take account of the thickness of the sides and base of the tank.

(1)

- (ii) How would your refinement affect your answer to part (a)?

(1)
 (Total 10 marks)

Q5.

In this question use $g = 9.81 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector $(40\mathbf{i} - 10\mathbf{j})$ metres.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that there are no resistance forces acting on the ball.

- (a) Find the speed of the ball when it is at a height of 3 metres above its initial position. (6)
- (b) State the speed of the ball when it is at its maximum height. (1)
- (c) Explain why the answer you found in part (b) may not be the actual speed of the ball when it is at its maximum height. (1)

(1)
 (Total 8 marks)

Q6.

A biologist is investigating the growth of a population of a species of rodent. The biologist proposes the model

$$N = \frac{500}{1 + 9e^{-\frac{t}{8}}}$$

for the number of rodents, N , in the population t weeks after the start of the investigation.

Use this model to answer the following questions.

- (a) (i) Find the size of the population at the start of the investigation. (1)
- (ii) Find the size of the population 24 weeks after the start of the investigation. Give your answer to the nearest whole number. (1)
- (iii) Find the number of weeks that it will take the population to reach 400. Give your answer in the form $t = r \ln s$, where r and s are integers. (3)

- (b) (i) Show that the rate of growth, $\frac{dN}{dt}$, is given by

$$\frac{dN}{dt} = \frac{N}{4000} (500 - N)$$

(4)

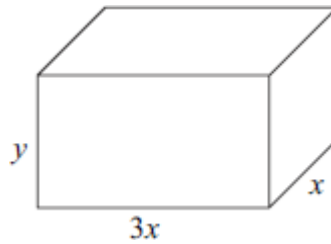
- (ii) The maximum rate of growth occurs after T weeks. Find the value of T .

(4)

(Total 13 marks)

Q7.

The diagram shows a solid cuboid with sides of lengths x cm, $3x$ cm and y cm.



The total surface area of the cuboid is 32 cm^2 .

- (a) (i) Show that $3x^2 + 4xy = 16$.

(2)

- (ii) Hence show that the volume, $V \text{ cm}^3$, of the cuboid is given by

$$V = 12x - \frac{9x^3}{4}$$

(2)

- (b) Find $\frac{dV}{dx}$.

(2)

- (c) (i) Verify that a stationary value of V occurs when $x = \frac{4}{3}$.

(2)

- (ii) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = \frac{4}{3}$.

(2)

(Total 10 marks)

Q8.

The volume, $V \text{ m}^3$, of water in a tank after time t seconds is given by

$$V = \frac{t^3}{4} - 3t + 5$$

- (a) Find $\frac{dV}{dt}$. (2)
- (b) (i) Find the rate of change of volume, in $\text{m}^3 \text{s}^{-1}$, when $t = 1$. (2)
- (ii) Hence determine, with a reason, whether the volume is increasing or decreasing when $t = 1$. (1)
- (c) (i) Find the positive value of t for which V has a stationary value. (3)
- (ii) Find $\frac{d^2V}{dt^2}$, and hence determine whether this stationary value is a maximum value or a minimum value. (3)
- (Total 11 marks)**

Q9.

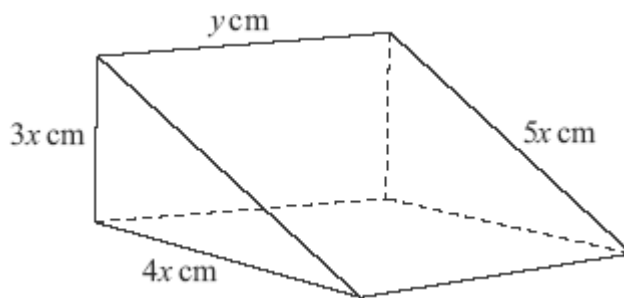
The depth of water, y metres, in a tank after time t hours is given by

$$y = \frac{1}{8}t^4 - 2t^2 + 4t, \quad 0 \leq t \leq 4$$

- (a) Find:
- (i) $\frac{dy}{dt}$. (3)
- (ii) $\frac{d^2y}{dt^2}$. (2)
- (b) Verify that y has a stationary value when $t = 2$ and determine whether it is a maximum value or a minimum value. (4)
- (c) (i) Find the rate of change of the depth of water, in metres per hour, when $t = 1$. (2)
- (ii) Hence determine, with a reason, whether the depth of water is increasing or decreasing when $t = 1$. (1)
- (Total 12 marks)**

Q10.

The diagram shows a block of wood in the shape of a prism with triangular cross-section. The end faces are right-angled triangles with sides of lengths $3x$ cm, $4x$ cm and $5x$ cm, and the length of the prism is y cm, as shown in the diagram.



The total surface area of the five faces is 144 cm^2 .

- (a) (i) Show that $xy + x^2 = 12$.

(3)

- (ii) Hence show that the volume of the block, $V \text{ cm}^3$, is given by

$$V = 72x - 6x^3$$

(2)

- (b) (i) Find $\frac{dV}{dx}$.

(2)

- (ii) Show that V has a stationary value when $x = 2$.

(2)

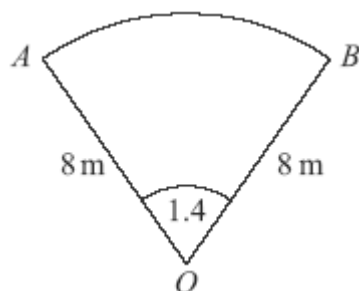
- (c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 2$.

(2)

(Total 11 marks)

Q11.

The diagram shows a sector OAB of a circle with centre O .



The radius of the circle is 8 m and the angle AOB is 1.4 radians.

- (a) Find the area of the sector OAB . (2)
- (b) (i) Find the perimeter of the sector OAB . (3)
- (ii) The perimeter of the sector OAB is equal to the circumference of a circle of radius x m. Calculate the value of x to three significant figures. (2)
- (Total 7 marks)**

Q12.

A model car moves so that its distance, x centimetres, from a fixed point O after time t seconds is given by

$$x = \frac{1}{2}t^4 - 20t^2 + 66t, \quad 0 \leq t \leq 4$$

- (a) Find:
- (i) $\frac{dx}{dt}$; (3)
- (ii) $\frac{d^2x}{dt^2}$. (2)
- (b) Verify that x has a stationary value when $t = 3$, and determine whether this stationary value is a maximum value or a minimum value. (4)
- (c) Find the rate of change of x with respect to t when $t = 1$. (2)
- (d) Determine whether the distance of the car from O is increasing or decreasing at the instant when $t = 2$. (2)
- (Total 13 marks)**

Q13.

Two numbers, x and y , are such that $3x + y = 9$, where $x \geq 0$ and $y \geq 0$.

It is given that $V = xy^2$.

(a) Show that $V = 81x - 54x^2 + 9x^3$. (2)

(b) (i) Show that $\frac{dV}{dx} = k(x^2 - 4x + 3)$, and state the value of the integer k . (4)

(ii) Hence find the two values of x for which $\frac{dV}{dx} = 0$. (2)

(c) Find $\frac{d^2V}{dx^2}$. (2)

(d) (i) Find the value of $\frac{d^2V}{dx^2}$ for each of the two values of x found in part (b)(ii). (1)

(ii) Hence determine the value of x for which V has a maximum value. (1)

(iii) Find the maximum value of V . (1)

(Total 13 marks)

Q14.

(a) The number of fish in a lake is decreasing. After t years, there are x fish in the lake. The rate of decrease of the number of fish is proportional to the number of fish currently in the lake.

(i) Formulate a differential equation, in the variables x and t and a constant of proportionality k , where $k > 0$, to model the rate at which the number of fish in the lake is decreasing. (2)

(ii) At a certain time, there were 20 000 fish in the lake and the rate of decrease was 500 fish per year. Find the value of k . (2)

(b) The equation

$$P = 2000 - Ae^{-0.05t}$$

is proposed as a model for the number of fish, P , in another lake, where t is the time in years and A is a positive constant.

On 1 January 2008, a biologist estimated that there were 700 fish in this lake.

- (i) Taking 1 January 2008 as $t = 0$, find the value of A .

(1)

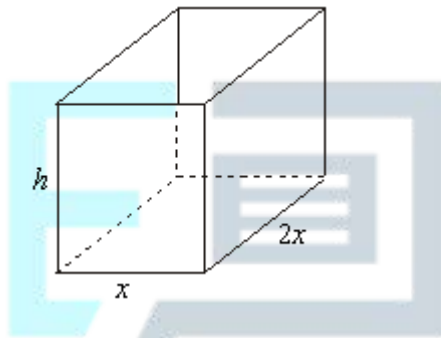
- (ii) Hence find the year during which, according to this model, the number of fish in this lake will first exceed 1900.

(4)

(Total 9 marks)

Q15.

The diagram shows an **open-topped** water tank with a horizontal rectangular base and four vertical faces. The base has width x metres and length $2x$ metres, and the height of the tank is h metres.



The combined internal surface area of the base and four vertical faces is 54 m^2 .

- (a) (i) Show that $x^2 + 3xh = 27$.

(2)

- (ii) Hence express h in terms of x .

(1)

- (iii) Hence show that the volume of water, $V \text{ m}^3$, that the tank can hold when full is given by

$$V = 18x - \frac{2x^3}{3}$$

(1)

- (b) (i) Find $\frac{dV}{dx}$.

(2)

- (ii) Verify that V has a stationary value when $x = 3$.

(2)

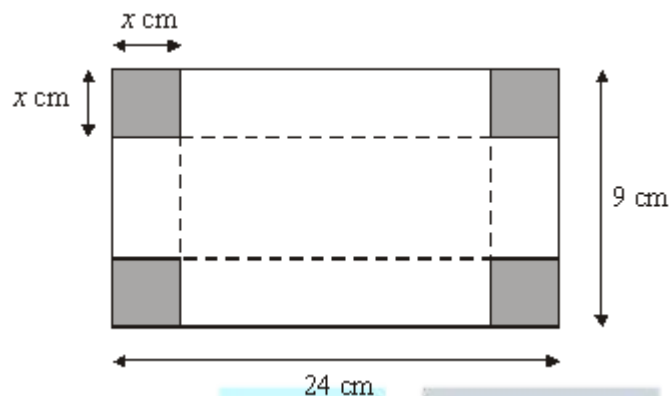
- (c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 3$.

(2)

(Total 10 marks)

Q16.

The diagram below shows a rectangular sheet of metal 24 cm by 9 cm.



A square of side x cm is cut from each corner and the metal is then folded along the broken lines to make an open box with a rectangular base and height x cm.

- (a) Show that the volume, V cm³, of liquid the box can hold is given by

$$V = 4x^3 - 66x^2 + 216x$$

(3)

- (b) (i) Find $\frac{dV}{dx}$.

(3)

- (ii) Show that any stationary values of V must occur when $x^2 - 11x + 18 = 0$

(2)

- (iii) Solve the equation $x^2 - 11x + 18 = 0$.

(2)

- (iv) Explain why there is only one value of x for which V is stationary.

(1)

- (c) (i) Find $\frac{d^2V}{dx^2}$.

(2)

- (ii) Hence determine whether the stationary value is a maximum or minimum.

(2)

(Total 15 marks)



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